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Research and application of smart streetlamp based on fuzzy control method

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Abstract

To meet the demand of the construction of smart city and make automatic adaption of streetlamp possible, a streetlamp control method based on fuzzy control theory is proposed. Firstly, the fuzzy control rules are designed to make the brightness of lamp node easy to adjust with the natural light dynamically. Secondly, a smart streetlamp system is designed by combining the wireless sensors network technology based on 6LoWPAN communication and visual program technology based on visual basic. Finally, the software and hardware test are conducted which achieves streetlamp adjust dynamically, remote control and data transmitted. Results shows: the lamps which are controlled by the fuzzy controller could adjust its brightness according to the environment smartly. The mentioned method can realize intelligent regulation and energy saving effect.

Keywords: Smart city; intelligent street lamp; Street lamp brightness; fuzzy control method; 6LoWPAN; wireless sensor network

1. Introduction

Street lamp, as the infrastructure of urban lighting, is an important part of the construction of smart city [1]. It is a research direction in the future to design ordinary street lamps to be able to adjust brightness "intelligently" according to environmental conditions. In order to achieve the purpose of automatic control and energy saving of street lamp brightness, NIU P.J. and others put forward the idea that the number of street lamps can be automatically adjusted according to the speed of vehicles on the road [2], LI S.F. and others put forward the idea of dynamic

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control of street lamps that "when people and vehicles arrive, the lights will be on, and when people and vehicles go, the lights will be off"[3], LIN H.J and others put forward the adaptive algorithm of street lamp light intensity based on neural network [4], QU Y.N, ZHANG B.C, etc. proposed the application of fuzzy PID algorithm to realize the control of street lamp [5-6].

In this paper, environmental protection and green lighting in the process of smart city construction, the sensor is used to collect the light and pollutant concentration around the street lamp in real time. According to the set fuzzy control rule algorithm, the brightness of the street lamp is dynamically and real-time regulated to realize the function that the brightness of the street lamp can be automatically adjust

2. Principle of fuzzy control

In the design, the output light intensity of the street lamp changes slowly with the change of the surrounding natural light, realizing the real-time adjustment of the output brightness of the street lamp, achieving a more user-friendly and energy-saving control effect [7-8], but the mathematical model of the controlled street lamp is difficult to achieve through system identification or the establishment of self attribute equation. Fuzzy control is a computer digital control technology based on fuzzy set theory, fuzzy language variable and fuzzy logic reasoning. It can achieve better control effect without knowing the specific mathematical model of the controlled object. The fuzzy control method does not need to establish the mathematical model of the controlled object, and can control the controlled object according to the fuzzy rules [9]. Therefore, in the control of street lamp brightness, the fuzzy controller and fuzzy control method can be used to realize the automatic adjustment of street lamp brightness with the natural light intensity. The structure of the designed fuzzy controller is shown in Figure 1.

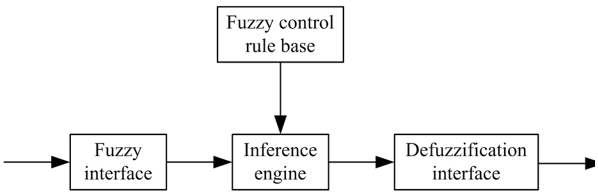

Fig.1. Structure Diagram of Fuzzy Controller

As can be seen from Figure 1, the fuzzy controller is divided into four parts: fuzzy interface, fuzzy control rule base, inference engine and fuzzy interface [10]. The function of fuzzy interface is to get a fuzzy vector and input it into the inference engine. The fuzzy control rule base is the most important part of the fuzzy controller and the core part of the fuzzy controller. Its function and function are equivalent to the corrector in the traditional control system. It needs to be debugged many times to get the appropriate fuzzy rules and provide the basis for the inference machine [11]. The function of the fuzzification interface is opposite to that of the fuzzification interface. It is to equivalent the fuzzy value or conclusion after the fuzzy reasoning to a clear value. The inference engine completes the final reasoning process according to the input of the fuzzy interface and the established fuzzy rules, and finally transfers them to the solution of the fuzzy interface, so as to realize the solution of the fuzzy relationship, and finally obtain the fuzzy control amount [12].

According to the above fuzzy control theory, in the design, the light intensity changes around the street lamp are collected by the photoresist, the deviation of the photoresist output voltage and the change rate of the deviation are taken as the input of the fuzzy controller, and the change of the voltage reflects the change of the brightness.

In actual operation, the deviation range of the sensor detection voltage is (- 1.3 ~ + 1.3) V, the deviation change rate range is (- 17.33 ~ + 17.33) V / s, the output range is (- 5 ~ + 5), the sampling period is 150ms, and the control value of the sensor voltage given by the design is 2V.

According to this, the luminance error E and its change rate EC of street lamps are set in the interval [- 1.3, + 1.3] and the interval [- 17.33,17.33] to change continuously. By discretizing them, a discrete set consisting of integers 0, 1, 2, 3, 4, 5 and 6 is formed, which is shown as follows:

$|e| = |ec| = \{0,1,2,3,4,5,6\}$

The corresponding fuzzy subsets of these two sets are set to:

$|e| = |ec| = \{$ NB、NM、NS、ZE、PS、PM、PB$\}$

The elements such as Nb, NM, NS, ZE, PS, PM and PBrepresent very dark, dark, dark, moderate, bright, bright and very bright for brightness [13]; for brightness change rate, they are rapidly darkening, rapidly darkening, gradually darkening, unchanged, gradually lightening, faster lightening and rapidly lightening.

The on domain of output is set as follow:

$P_o = \{-3, -2, -1, 0, 1, 2, 3\}$

The fuzzy subset is also set to:

$|e| = |ec| = \{$ NB、NM、NS、ZE、PS、PM、PB$\}$

At this time, the fuzzy process of input and output is completed. Because the change rate of resistance is large when the light of photosensitive resistance is weak, and the change rate of resistance is small when the light is strong. According to this feature, when setting the membership function of deviation and deviation change rate, the part with smaller brightness is denser, and the part with larger brightness is looser. The fuzzy rules are shown in Table 1, and the function of deviation and deviation change rate is shown in Figure 2.

Table 1. Fuzzy Rule Table(e: Deviation, de: Variation rate of deviation).

de\e	NB	NM	NS	ZE	PS	PM	PB
NB	PB	PB	PB	0	0	0	0
NM	PB	PB	PB	PM	PS	0	0
NS	PB	PB	PM	PS	ZE	NS	NM
ZE	PB	PM	PS	ZE	NS	NM	NB
PS	0	0	ZE	NS	NM	NB	NB
PM	0	0	0	NM	NB	NB	NB
PB	0	0	0	0	NB	NB	NB

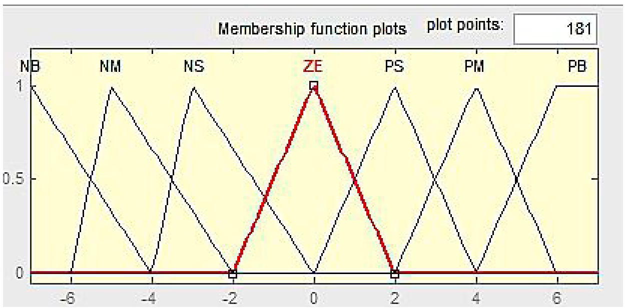

Fig. 2. Subordinate Degree Function of Deviation and Deviation Change Rate

In order to facilitate the calculation, the input data is magnified 100 times and then fuzzified. The scale factors obtained are: deviation (7 / 130), deviation change rate (7 / 1733), and output (7 / 5). Because the general process of solving fuzzy is too complicated, it affects the efficiency of the processor, and then affects the real-time performance of the whole system. Therefore, the fuzzy output data corresponding to each membership degree is calculated by MATLAB software, as shown in Table 2, and the contents of the table are written into an array of 15 * 15. The maximum minimum method and the weighted average method are selected as the calculation rules. When the fuzzy process is solved, the corresponding output can be obtained quickly by searching the array, which avoids the tedious calculation and ensures the control accuracy as well as the real-time performance.

Table 2. Fuzzy Output Query Table (e: Deviation, de: Variation rate of deviation).

de\e	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
-7	6	6	6	6	6	6	6	0	0	0	0	0	0	0	0
-6	6	6	6	6	6	6	6	0	0	0	0	0	0	0	0
-5	6	6	6	6	6	6	5	4	3	2	2	0	0	0	0
-4	6	6	6	6	6	6	5	4	3	2	2	0	0	0	0
-3	6	6	6	6	4	4	3	3	2	1	0	-2	-3	-4	-4
-2	6	6	6	6	4	4	3	2	1	0	-1	-2	-3	-4	-4
-1	6	6	5	5	3	3	2	1	0	-1	-2	-3	-4	-5	-5
0	6	6	5	4	3	2	1	0	-1	-2	-3	-4	-5	-6	-6
1	6	6	5	4	2	1	0	-1	-2	-3	-4	-5	-5	-6	-6
2	0	0	0	0	0	0	-1	-2	-3	-4	-5	-6	-6	-6	-6
3	0	0	0	0	0	0	-2	-3	-4	-5	-5	-6	-6	-6	-6
4	0	0	0	0	0	0	-4	-4	-5	-6	-6	-6	-6	-6	-6
5	0	0	0	0	0	0	-4	-4	-5	-6	-6	-6	-6	-6	-6
6	0	0	0	0	0	0	0	0	-6	-6	-6	-6	-6	-6	-6
7	0	0	0	0	0	0	0	0	-6	-6	-6	-6	-6	-6	-6

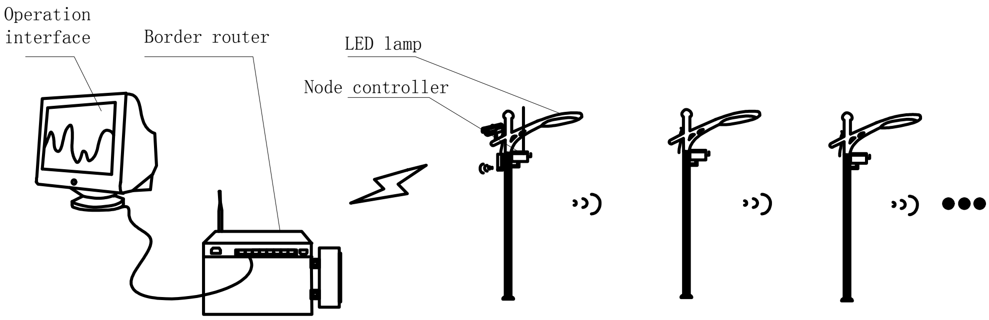

Fig. 3. Intelligent Street Lamp Overall Design Scheme

3. Intelligent street lamp system design

3.1. Overall scheme design

The overall design scheme of intelligent street lamp based on fuzzy control theory is shown in Figure 3. The scheme is composed of three parts: upper computer operation interface, street lamp control station and intelligent street lamp node. Among them, street lamp control station mainly includes wireless bridge and border router; intelligent street lamp node mainly includes wireless bridge, LED street lamp, node controller, etc.; upper computer operation interface realizes remote control of street lamp and remote transmission of street lamp data through visualization software.

3.2. Node of smart streetlamp

Intelligent street lamp node is the core part of the whole scheme design, which can be divided into two parts: data acquisition and data transmission. The hardware design is shown in Figure 4.

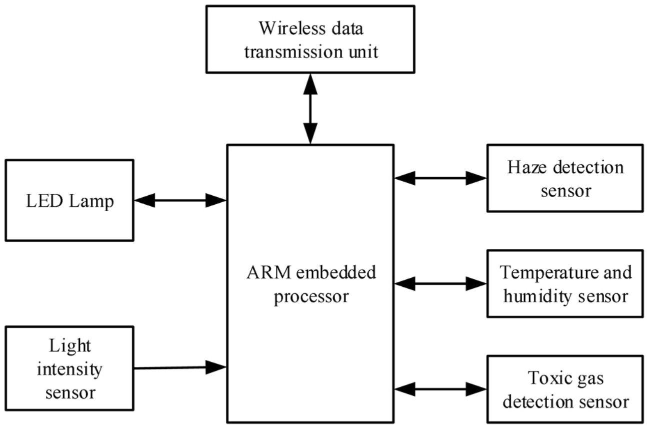

Fig. 4. Intelligent Street Lamp Node Design Block Diagram

(1) The data acquisition part can read the current electric current, voltage, brightness, temperature, humidity, PM2.5, toxic gas, light intensity and video image of street lamp, and realize the real-time adjustment of street lamp brightness. The data transmission part is to realize the remote wireless transmission of the collected data and video. Wireless data transmission adopts IPv6 over low power wireless personal area network communication protocol [14]. The protocol is based on the TCP / IP network architecture. Without any protocol conversion, it can realize the seamless connection with the Internet and simplify the network topology.

(2) ARM embedded processor adopts a 32-bit embedded processor STM32F103 chip with low power consumption and high performance [15]. The software of intelligent street lamp node is written in C language. In the process of software design, Contiki operating system is transplanted, and the program is designed based on the realization of user thread function under the operating system. The algorithm flow of automatic adjustment of street lamp brightness based on fuzzy control theory is shown in Figure 5. As can be seen from Figure 5, when the intelligent street lamp node receives the automatic adjustment instruction from the upper computer operation interface, the control program will automatically adjust the street lamp brightness according to the collected sensor data and the set fuzzy rules, so as to achieve the purpose of intelligent control and energy saving of street lamp brightness.

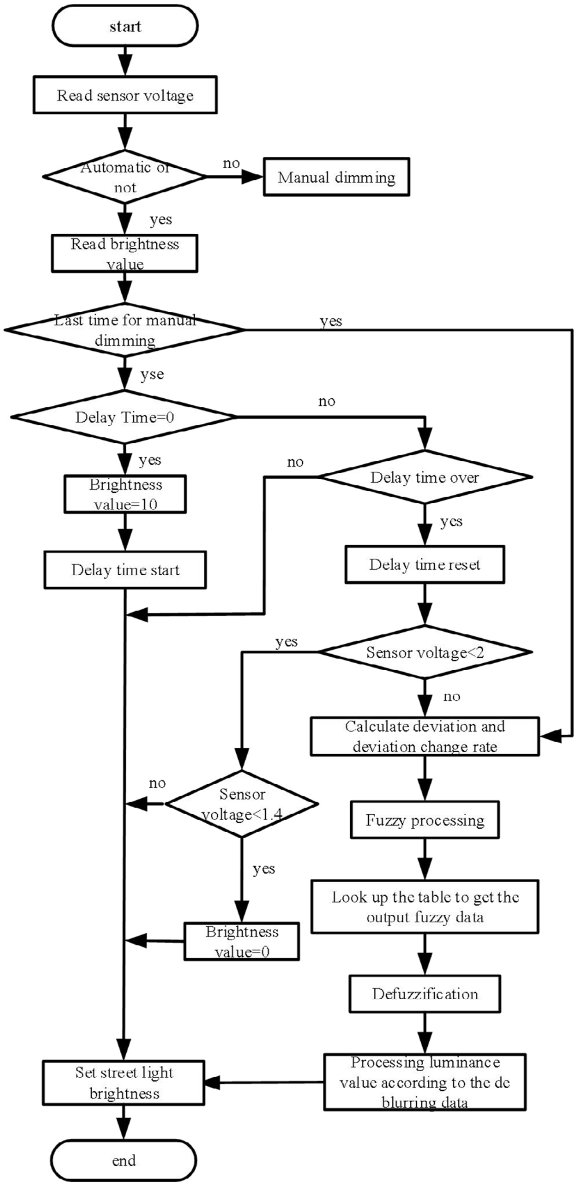

Fig. 5. Flow chart of intelligent street lamp brightness automatic adjustment

In order to realize the remote control of street lamp and make it display the control effect directly, the system also completed the design of the upper computer operation interface. The interface is compiled by VB (Visual Basic 6.0) program and visual programming technology. It is mainly divided into three parts: network communication setting, street lamp data viewing and street lamp brightness control. The intelligent street lamp node adjusts the output brightness of the street lamp in real time according to the fuzzy control rules.

4. Experiment and result analysis

The design of the system finally realizes the intelligent control and data collection function of street lamp. The designed intelligent street lamp controller and test site are shown in Figure 6. Change the light intensity around the street lamp so that the light conditions correspond to 7 levels of very dark, dark, dark, moderate, light, light and very bright. Test the automatic dimming effect of the street lamp. The test results are shown in Table 3.

Fig. 6. Intelligent Street Lamp Node Physical and Test Map

From the test results in Table 3, it can be seen that the street lamp designed with the fuzzy control theory can adjust the working current and voltage and the brightness of its output "intelligently" according to the lighting conditions around it. When the light intensity around the street lamp increases gradually, the working current of the street lamp decreases gradually, and its brightness also decreases gradually. If the light intensity around the street lamp is very strong, the street lamp will automatically turn off the working current, and reduce the output voltage, so as to realize the intelligent adjustment of street lamp brightness and the effect of energy saving.

Table 3. Test results of street lamp automatic dimming data.

Light conditions	Dust density (ug/m3)	lamp voltage (V)	lamp current (mA)	lamp brightness (%)
Very dark	18.57	114	580	100
darker	19.72	114	420	70
dim	18.43	114	240	40
moderate	18.49	114	100	21
partial brightness	19.52	114	82	16
brighte	20.02	114	61	10
very bright	18.82	78	0	0

5. Conclusion

In this paper, the theory of fuzzy control is deeply studied, and a set of fuzzy control rules for the function of automatic adjustment of street lamp brightness with natural light is designed. The results of MATLAB software simulation are applied to the control program of intelligent street lamp node. After the individual debugging of each functional unit and the overall debugging of the system, the remote monitoring of street lamp and the intelligent adjustment of street lamp brightness are realized. Compared with the traditional control method of street lamp, the automatic dimming function of street lamp using fuzzy control algorithm can not only make the lighting service of street lamp more scientific and reasonable, but also achieve more energy-saving effect of urban road lighting. In addition, the design of intelligent street lamp system based on 6LoWPAN wireless communication protocol can also collect haze, temperature and humidity, toxic gas in the city and provide video monitoring data, which provides the channel and interface of basic information collection for the construction of intelligent city and lays a good foundation for the operation and management of intelligent city. At the same time, it also provides some reference value for the promotion and application of 5G technology in the construction of smart city.

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